

Vishal Ludhani

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SKILLS

- **Languages:** C, C++, Java, Python, JavaScript
- **Web Technologies:** React.js, Node.js, Express.js, HTML, CSS
- **Databases:** MongoDB, SQL
- **Core CS:** Data Structures & Algorithms, OOP, DBMS, Operating Systems
- **Tools:** Git, GitHub, VS Code, Postman

PROJECTS

Real-Time Collaborative Code Editor

(MERN + Socket.io)

- Engineered a scalable real-time collaborative platform enabling concurrent multi-user editing with room-based architecture
- Achieved low-latency code synchronization using Socket.io (WebSockets), ensuring near real-time updates across clients
- Designed modular backend (Node.js, Express, MongoDB) following MVC architecture for maintainability and scalability
- Developed RESTful APIs for room management, session handling, and collaborative state synchronization
- Integrated Judge0 API for multi-language code execution with efficient request-response handling
- Implemented live cursor tracking, user presence, and in-room chat to enhance collaborative experience
- Ensured data consistency and handled concurrent updates across multiple users

Expense Analytics System

(MERN Stack)

- Built full-stack expense tracking system with analytical dashboards for categorized financial insights
- Designed RESTful APIs with modular architecture improving scalability and code maintainability
- Implemented secure authentication using JWT with HttpOnly cookies to prevent XSS attacks
- Optimized MongoDB queries and aggregation pipelines for efficient data retrieval

Role-Based Blog Website

(MERN Stack)

- Developed blogging platform with Role-Based Access Control (RBAC) for secure resource authorization
- Implemented protected API endpoints and backend authorization middleware
- Designed normalized MongoDB schemas ensuring data integrity and scalability

ACHIEVEMENTS

- Winner, Vogue AI Challenge (GDSC) – Developed AI prompt optimization model improving output accuracy
- LeetCode Rating: 1365 | Solved 240+ Data Structures & Algorithms problems

EDUCATION

Anurag University

B.Tech in Computer Science Engineering

Relevant Coursework: Data Structures, OOP, DBMS, Operating Systems, Computer Networks

Expected 2027

CGPA: 8.56